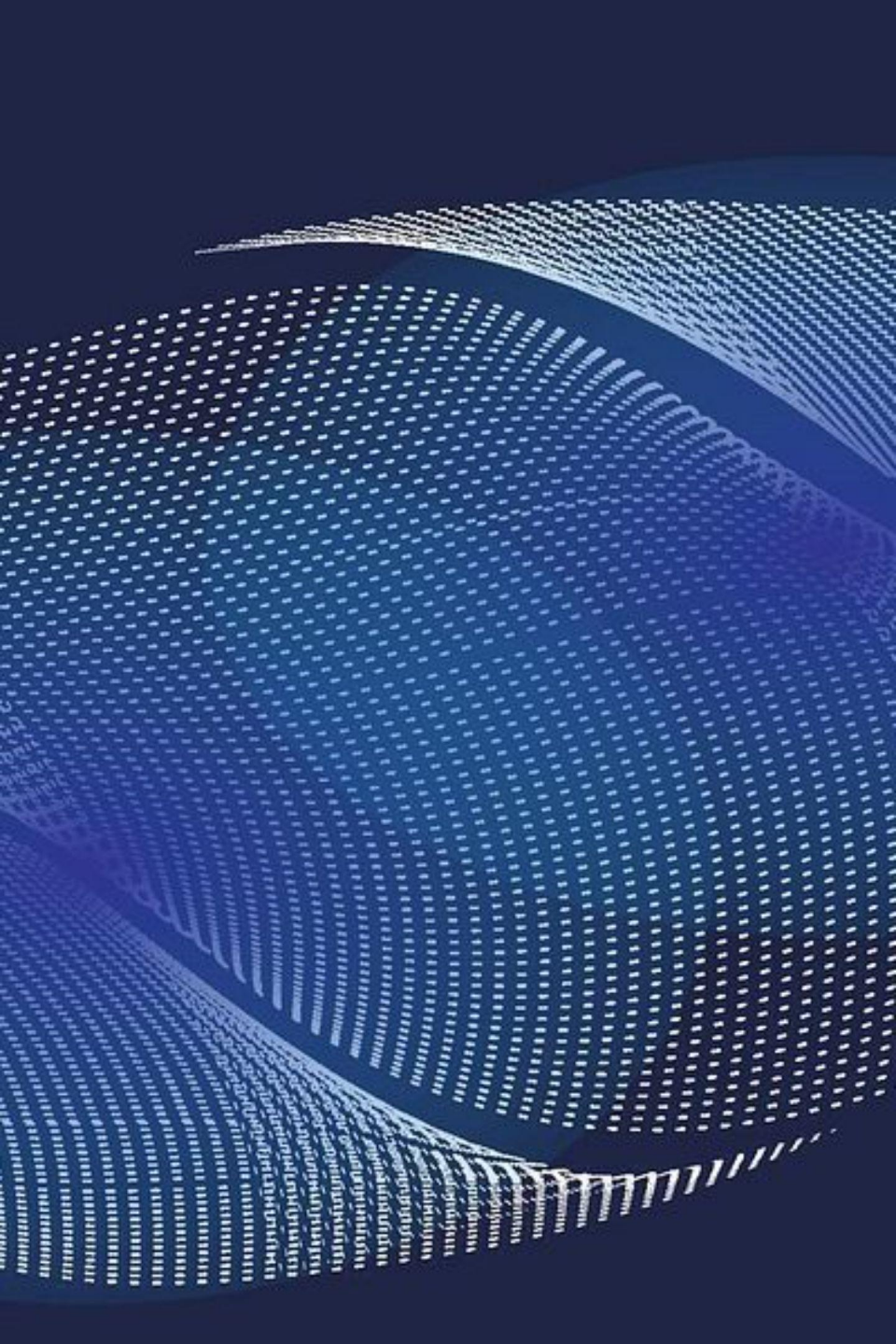


An abstract graphic on the left side of the page featuring a series of concentric, overlapping wave-like patterns in shades of blue and white, creating a sense of depth and movement.

SOUND SENSOR

An Intelligent Acoustic Sensing Component

Student Name

Nikhil Kumar Reddy Kadiveti

Register No

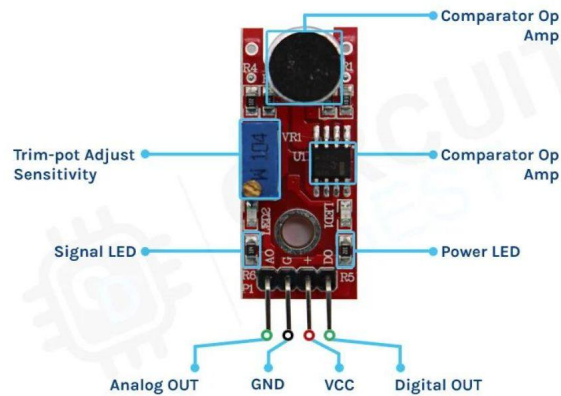
192425031

Course / Subject

MLA0104

SOUND SENSOR: Image & Components

Original sensor image with labelled components (use provided image below). Explanations match the labels visible on the board.



Comparator Op-Amp

Compares incoming microphone signal to a threshold and drives the digital output when the threshold is exceeded.



Trim-Pot – Adjust Sensitivity

Adjusts the comparator threshold so the sensor triggers at the desired sound level.



Signal LED

Provides visual indication when sound is detected above the set threshold (tied to the comparator output).



Power LED

Indicates that the sensor board is powered (VCC applied and circuit active).



Analog OUT

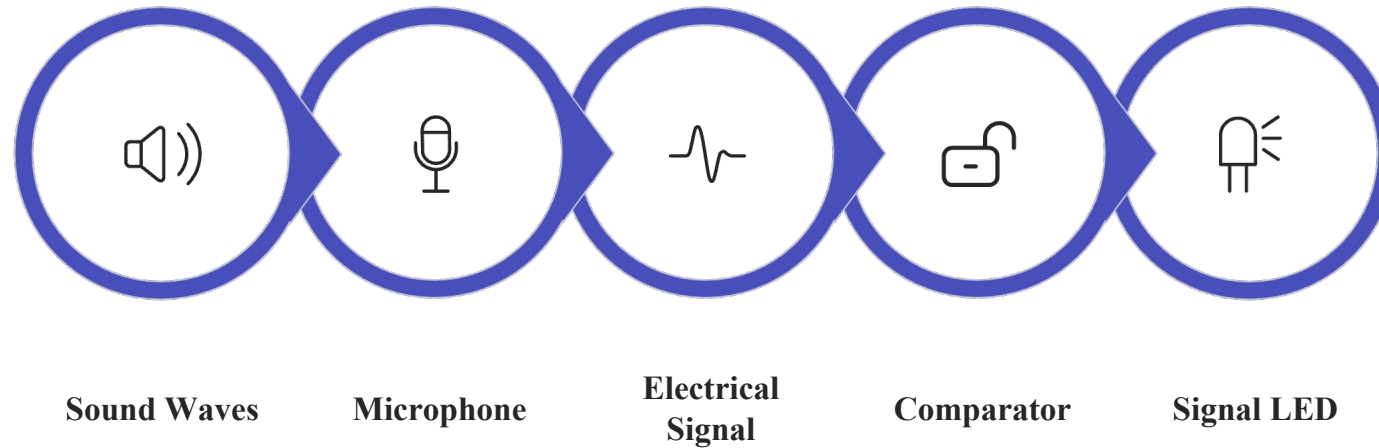
Continuous analog voltage representing the raw microphone signal amplitude over time.



Digital OUT

Binary output (HIGH/LOW) driven by the comparator when the signal crosses the threshold set by the trim-pot.

How the Components Work



Process summary:

1. Sound waves reach the microphone element; the microphone converts acoustic pressure into a small voltage waveform.
2. The raw waveform appears at the board and is available at Analog OUT as a continuous voltage proportional to amplitude.
3. The Comparator Op-Amp monitors the waveform and compares it against the threshold set by the Trim-Pot.
4. When the waveform exceeds the threshold, the comparator output switches: Digital OUT goes HIGH and the Signal LED lights; otherwise Digital OUT is LOW.

Sensitivity Trim-Pot

Adjusts the comparator reference voltage—turning it increases/decreases the sound level required to trigger the digital output.

VCC & GND

VCC supplies operating voltage to the microphone preamp and comparator; GND is the common reference. Stable supply ensures reliable thresholds and LED indicators.

Analog OUT vs Digital OUT

Analog OUT = continuous waveform for measurement or logging.
Digital OUT = thresholded event signal (easy to read by microcontrollers or trigger circuits).

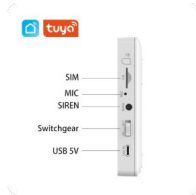
Where It Is Used & How

Practical applications and how the sound sensor is applied in each case.



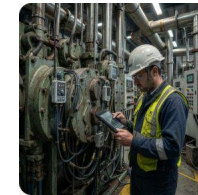
Smart Home

Used for clap or sound-based triggers (lights, appliances). Digital OUT connects to a controller that interprets short audio events as commands.



Security

Detects events like breaking glass or loud disturbances. Digital OUT flags events for alarm systems; Analog OUT can be recorded for verification.



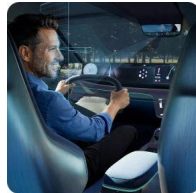
Industrial Monitoring

Monitors abnormal machine sounds. Analog OUT enables waveform analysis; thresholded Digital OUT triggers alerts during anomalies.



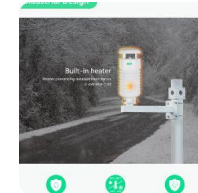
Healthcare

Acoustic monitoring for cough, respiration, or equipment alerts. Analog data supports signal processing; Digital events can flag incidents in real time.



Automotive

Used for in-cabin noise detection, crash or impact sensing supplements other sensors; Digital OUT integrates with vehicle control logic for immediate responses.



Environmental Noise Monitoring

Measures ambient sound levels for compliance or studies. Analog OUT feeds to data loggers; thresholds on Digital OUT can mark exceedance events.